PHYTOTHERAPY RESEARCH
Phytother. Res. **22**, 442–449 (2008)
Published online 12 March 2008 in Wiley InterScience
(www.interscience.wiley.com) **DOI**: 10.1002/ptr.2325

Effects of a Herbal Gel containing Carvacrol and Chalcones on Alveolar Bone Resorption in Rats on Experimental Periodontitis

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Carvacrol and dimeric chalcones are the respective bioactive components of *Lippia sidoides* and *Myracrodruon urundeuva*, popular medicinal plants of Northeastern Brazil with proven antimicrobial and antiinflammatory properties. Periodontal disease is associated with inflammation and microbiological proliferation, thus the study aimed to investigate the effect of a topical gel based on carvacrol and chalcones in the experimental periodontal disease (EPD) in rats. Animals were treated with carvacrol and/or chalcones gel, immediately after EPD induction, three times a day for 11 days. Appropriate controls were included in the study. Animals were weighed daily. They were killed on day 11, the mandibles dissected and alveolar bone loss was measured. The periodontium were examined at histopathology and the neutrophil influx into the gingiva was assayed using myeloperoxidase activity. The bacterial flora were assessed through culture of the gingival tissue. Alveolar bone loss was significantly ($p < 0.05$) inhibited by combined carvacrol and chalcones gel, compared with the vehicle and non-treated groups. The treatment with the combined gel reduced tissue lesion at histopathology, decreased myeloperoxidase activity in gingival tissue and inhibited the growth of oral microorganisms as well as the weight loss. Carvacrol and chalcones combination gel has a beneficial effect upon EPD in this model.

Keywords: chalcones; carvacrol; antimicrobial activity; periodontitis; antiinflammatory activity.

INTRODUCTION

Several plant-derived compounds have been evaluated for their antiplaque activity on periodontal bacteria such as phenolic-related essential oils and herbal extracts dental gels (Fine *et al.*, 2000; Botelho *et al.*, 2007c). Although antibacterial effect of essential oil derivatives is well acknowledged, the mode of treatment and delivery system for maximal effectiveness is not yet fully clear (Pai *et al.*, 2004). From this perspective, the use of antimicrobial compounds in gel vehicle may be considered a useful adjunct to oral hygiene. Previous studies demonstrated that an essential oil containing thymol and carvacrol showed antiplaque effectiveness on periodontal bacteria (Osawa *et al.*, 1990). Unfortunately, in third world countries, the great majority of the commercially available antiplaque agents are out of reach for patients. Hence, there is a need to discover new effective, safe and relatively inexpensive agents.

Lippia sidoides (family *Verbenaceae*) is a medicinal plant that has been used in traditional Brazilian medicine for a variety of antifungal and antimicrobial purposes (Lacoste *et al.*, 1996). Biochemical data have shown that oils and extracts from *L. sidoides* are rich in thymol and carvacrol compounds, known for their antimicrobial properties (Lemos *et al.*, 1990). An initial short-term controlled animal study indicated that *L. sidoides* E.O-based mouthrinse reduced the severity of gingival inflammation, bacterial plaque and histological inflammatory infiltrate in dogs (Girao *et al.*, 2003). Its antimicrobial activity *in vitro* against oral pathogens related to caries and oral diseases was described previously (Botelho *et al.*, 2007b).

Myracrodruon urundeuva (Engl.), popularly known as 'aroeira do sertao', is a medicinal plant traditionally used for treating bleeding gums and in gynecological disorders (Viana *et al.*, 2003; Monteiro *et al.*, 2006). Previous studies have shown the analgesic and antiinflammatory effects of tannin and chalcones enriched fractions (CEF) isolated from this plant in experimental models of inflammation, such as carrageenan-induced paw edema, neutrophil migration to peritoneal cavity and hemorrhagic cystitis induced by cyclophosphamide (Viana *et al.*, 1997; Viana *et al.*, 2003). Analgesic, antiulcer, hepatoprotective, antidiarrheal and wound healing properties have been also reported (Viana *et al.*, 1997). The essential oil containing mouthrinses of *Lippia sidoides* plant origin are available in the market for the treatment of oral infections (Matos, 1989). Thymol and

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Contract/grant sponsor: FUNCAP (Fundação Cearense de Apoio ao Desenvolvimento Científico e Tecnológico); contract/grant number: 9131/06.

Received 11 March 2007
Revised 12 July 2007
Accepted 18 July 2007

carvacrol show a good antibacterial efficacy against the microbes responsible for the oral infections (Osawa *et al.*, 1990; Botelho *et al.*, 2007b). Based on the assumption of obtaining better efficacy, the present study was aimed to develop a mucoadhesive gel containing carvacrol from *Lippia sidoides* essential oil and CEF from the stem bark of *M. urundeuva* and to evaluate its antiinflammatory and antimicrobiological effect in rats on experimental periodontitis.

MATERIALS AND METHODS

Plant material. Initially, Federal University of Ceará Ethical Committee (#551/04; #228/06) authorized the collection of the plants as follows: the leaves of *Lippia sidoides* were collected in August 2005 at the Medicinal Plants Garden Prof. Francisco José Abreu Matos and the *Myracrodruon urundeuva* ground bark was collected from the outskirts of the city of Fortaleza, state of Ceará, Brazil, and the voucher specimens (25.149 and 14.999, respectively) were deposited at the Prisco Bezerra Herbarium, after authentication by botanists of the, Department of Biology, Federal University of Ceará.

***Lippia sidoides* essential oil gas chromatography–mass spectrum analysis.** The essential oil from *L. sidoides* was obtained by hydrodistillation of the fresh leaves (300 g) using a Clevenger-type glass apparatus (Craveiro *et al.*, 1976) for 2 h. The oil was dried over anhydrous sodium sulphate, yielding 1.0%. After extraction, the volume of essential oil obtained was measured and conditioned in hermetically sealed glass containers with rubber lids, covered with aluminum foil to protect the contents from light and kept under refrigeration at 8 °C until used. The chemical composition of the essential oil was determined at the Technological Development Center (PADETEC) at the Federal University of Ceará by Gas Chromatography coupled to mass spectroscopy (GC-MS) using a Hewlett-Packard 5971 GC/MS apparatus under the following conditions: polydimethylsiloxane DB-1 fused silica capillary column (30 m × 0.25 mm), with film thickness 0.10 µm; the carrier gas was helium (1 mL/min), the injector temperature was 250 °C whilst the detector temperature was 200 °C. The column temperature ranged from 35 to 180 °C/min, at 4 °C V/min, and then from 180 to 280 °C, at 20 °C V/min; mass spectra were obtained by electronic impact 70 eV. The identification of the constituents was performed by a computer-based library search, with retention indices and visual interpretation of the mass spectra (Alencar *et al.*, 1984).

Chalcone enriched fraction (CEF) from *M. urundeuva*. The chalcone enriched fraction (CEF) was obtained from the ethyl acetate extract prepared with 5 kg of *M. urundeuva* ground stem bark as described previously (Viana *et al.*, 2003). The ethyl acetate extract was submitted to a treatment with hexane in a soxhlet apparatus, in order to remove lipids. The extract was then chromatographed in a silica gel column (5.0 cm diameter) and eluted with chloroform and mixtures of chloroform/acetone in the following proportions: 9:1, 8:2, 7:3, 1:1, followed by acetone and finally by methyl alcohol. The isolation of the chalcone type fraction compounds was done with a corn starch as the column support and elution with a mixture of dichloromethane/methanol (95:5). Through chromatographic procedure, three compounds were isolated from that fraction (CEF). The analytical procedure used for structural identification of these compounds involved RMN ^{1}H and RMN ^{13}C (HBBD and DEPT (*) = 13.5), with the contribution from bidimensional spectra of a homonuclear correlation of hydrogen (^{1}H, ^{1}H-COSY), as well as a heteronuclear correlation of hydrogen and carbon H. ^{13}C-COSY $*\int_{C8} \int n = \int$ (HMQC) and $n = 2$ and $_{C8} = 3$, corresponding to HMBC and ^{1}H, ^{1}H-NOESY, respectively. The precise attribution of chemical displacements of carbon and hydrogen atoms revealed, for the first time, that the CEF is composed of three dimeric chalcones, named *urundeuvina* A, B and C (Fig. 1).

Gel preparation. The gels containing chalcones (600 µg/g) and carvacrol (300 µg/g) or their combination were prepared at the Faculty of Pharmacy, Federal University of Ceará. The flask with active ingredients of the both plants was kept in the dark to avoid effect of light. For the preparation of both herbal gels, carbopol-94O (BF Goodrich Co., Cleveland) 2% (w/v) was used by mechanical dispersion in distilled water under vigorously agitation. For the preparation of chalcones enriched fraction (CEF) 600 µg/g gel 10 mL of carbopol-94O, 6 mg of isolated chalcones was used, being neutralized until pH 6.0 with triethanolamine. The carvacrol (300 µg/g) gel was prepared with 10 mL of carbopol-94O with 3 mg of carvacrol, 5 mL polysorbate 80, being neutralized to pH 6.0 with triethanolamine. The combined gel was prepared by the combination of these two techniques. (Brazilian Pharmacopeia). The doxicicline (100 µg/g) gel was purchased from Evidence Pharmaceuticals, Fortaleza, Brazil. The gels were stored in white polyethylene containers kept hermetically sealed under refrigeration at 8 °C until used.

Stability study to evaluate the consistency of the gel over a period of 2 months was conducted by keeping the formulation at different conditions (4 °C, 37 °C and room temperature) and measuring the viscosity of the gel formulation at regular intervals. The viscosity was measured by Brooke-field synchrolectric viscometer. The TD bar spindle of LV series was employed for the measurement. The study indicated that the viscosity of the carbopol gel did not change significantly throughout the stability period in the specified conditions.

Animals. One hundred and eight male Wistar rats (160–200 g) from our own animal house were housed in temperature-controlled rooms and received water and food *ad libitum*. All experiments were conducted in accordance with local guidelines on the welfare of experimental animals and with the approval of the Committee of Ethics in Animal Research of the Federal University of Ceará (#10/05).

Induction of experimental periodontal disease (EPD). A sterilized nylon (000) thread ligature was placed around the cervix of the second left upper molar of rats anesthetized with 10% chloral hydrate (400 mg/kg, i.p.), as described elsewhere (Menezes *et al.*, 2005). The ligature was knotted on the buccal side of the tooth, resulting in subgingival position palatinally and in

DOI: 10.1002/ptr

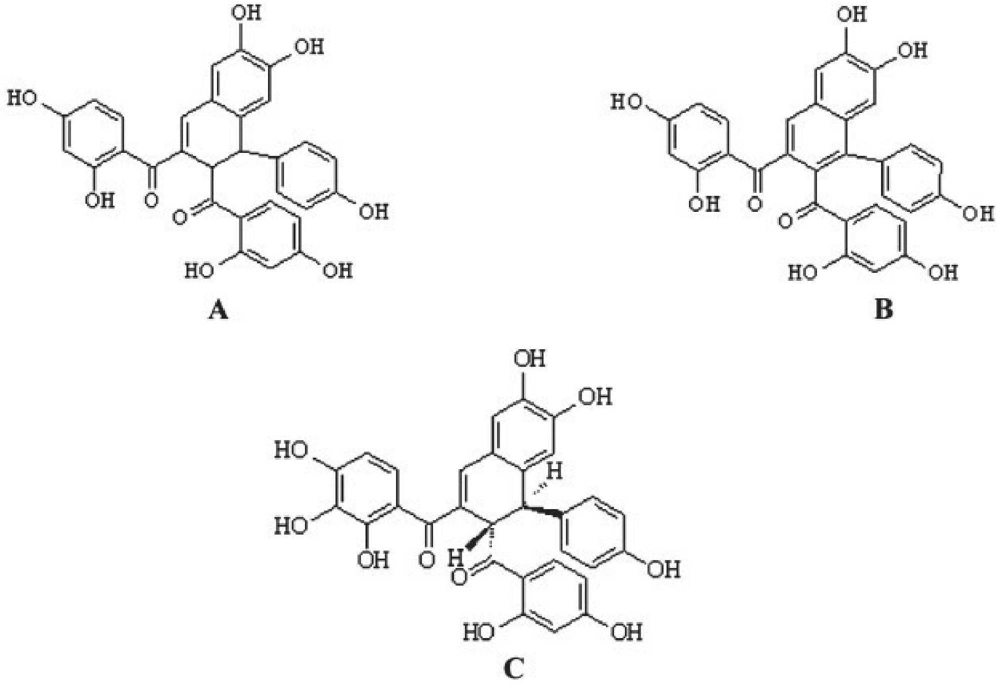

Figure 1. Chemical structures of chalcones enriched fraction (CEF) *urundevinas* A, B and C isolated from *M. urundeuva* locally applied as a dental gel.

supragingival position buccally. The contralateral right side was used as the unligated control (Menezes *et al.*, 2005). Animals were weighed daily.

Drug treatments. The experiments were performed utilizing six animals per group. The experimental groups were divided into: naive group (animals non-treated and not subjected to EPD); non-treated group (animals subjected to EPD which received no treatment); vehicle-treated group (animals subjected to EPD and treated with topical vehicle gel); CEF group (animals subjected to EPD and treated with topical chalcones 600 μg/g gel; CAR group (animals subjected to EPD and treated with topical carvacrol 300 μg/g gel; CAR + CEF group (animals subjected to EPD and treated with topical chalcones 600 μg/g and Carvacrol 300 μg/g gel combined based gel) and finally the DOX treated group (positive control) (animals subjected to EPD and treated with topical doxycycline 100 μg/g gel); The topical treatment with the gel was performed immediately after the surgical procedure and three times/day until killing on day 11.

Measurements of alveolar bone loss. The animals were killed on day 11 of periodontitis induction, and had their maxillae excised and fixed in 10% neutral formalin. Both maxillary halves were then defleshed and stained with aqueous methylene blue (1%) in order to differentiate bone from teeth. The horizontal alveolar bone loss, the distance between the cusp tip and the alveolar bone, was measured using a modification of the method of Crawford *et al.* (1978) as described by Samejima *et al.* (1990). Measurements were made along the axis of each root of the first (three roots), second and third molar teeth (two roots). The total alveolar bone loss was obtained by taking the sum of the recordings from the buccal tooth surface and subtracting the value of the right maxilla (unlighted control) from the left, in millimeters (Leitão *et al.*, 2005).

Histopathological analysis. After killing by anesthesia, the animals had their maxillae excised. The specimens were fixed in 10% neutral buffered formalin and demineralized in 7% nitric acid. These specimens were then dehydrated, embedded in paraffin, and sectioned along the molars in a mesio-distal plane, for hematoxylin and eosin staining. Sections of 6 μm thickness, which included the roots of the first and second molars, were used. The areas between the first and second molars, where the ligature was placed, were analysed under light microscopy using a 0–3 score grade, considering the inflammatory cell influx, and alveolar bone and cementum integrity, as described previously (Menezes *et al.*, 2005): Score 0: absence or only a discrete cellular infiltration (inflammatory cell infiltration is sparse and restricted to the region of the marginal gingival), preserved alveolar process and cementum. Score 1: moderate cellular infiltration (inflammatory cellular infiltration present all over the insert gingival), some but minor alveolar process resorption and intact cementum. Score 2: accentuated cellular infiltration (inflammatory cellular infiltration present in both gingival and periodontal ligament), accentuated degradation of the alveolar process, and partial destruction of cementum. Score 3: accentuated cellular infiltrate, complete resorption of the alveolar process and severe destruction of cementum.

Measurement of myeloperoxidase activity. The myeloperoxidase (MPO) activity in the gingival tissue, collected at 6 h after EPD induction was determined as a measurement of neutrophil accumulation. A spectrophotometric assay was used to measure MPO activity, as described previously (Bradley *et al.*, 1982). The buccal gingiva surrounding the upper left molars were removed and stored at –70 °C. The material was suspended in 0.5% hexadecyltrimethylammonium bromide (HTAB) in 50 mM potassium phosphate buffer, pH 6.0, to solubilize MPO. After homogenized in an ice bath

DOI: 10.1002/ptr

(15 s), the samples were freeze-thawed twice. Additional buffer was added to the test tube to reach 400 μL of buffer per 15 mg of tissue for 12 min. After centrifuging (1000 **g**/12 min), 0.1 mL of the supernatant was added to 2 mL phosphate buffer 50 mM, pH 6.0, containing 0.167 mg/mL o-dianosidine dihydrochloride, distilled water and 0.0005% hydrogen peroxide to give a final volume of 2.1 mL per tube. The absorbance was measured spectrophotometrically (460 nm). One unit of activity was defined as that degrading 1 μmol of peroxide/min at 25 °C. The results are expressed in myeloperoxidase units/mL. Staining of smears for MPO activity was performed by the method of Kaplow (1986).

Microbiological analysis. The buccal gingiva surrounding the upper left molars were removed and placed in 0.3 mL of brain heart infusion broth (BHI). The total transfer time to the microbiology laboratory was less than 1 h. Immediately after, the collected fragment was homogenized and plated in dilutions 1:100 and 1:1000 into blood agar (brain heart infusion agar supplemented with 5% defibrinated sheep blood and henin/menadione 10 μg/mL). The microaerophilic environment was obtained using commercially available kits. In order to determine the antimicrobial activity of Car and CEF gels, the samples counts of microorganisms (colonies forming units), were done from gingival tissue of rats subjected to periodontitis. Twenty four hours later the Petri dishes were counted in each group and compared with the groups used as controls in the experiments.

Statistical analysis. The data are presented as the mean ± SEM or as the medians, where appropriate. A univariate analysis of variance (ANOVA) followed by Bonferroni's test was used to compare means, and the Kruskal-Wallis test was used to compare medians. A probability value of $p < 0.05$ was considered to indicate significant differences.

RESULTS AND DISCUSSION

The present study showed that the local application of a dental gel based on chalcones enriched fractions (CEF) isolated from *Myracrodruon urundeuva* combined with carvacrol isolated from *Lippia sidoides* essential oil, plays a positive role on the bone loss associated with experimental periodontal disease. This effect was associated with a reduction of the inflammatory reaction and antimicrobial activity. In accordance with our data, a clinical trial has demonstrated that a mouth-rinse based on *Lippia sidoides* essential oil decreased the clinical scores and histopathological aspect of gingival health in dogs compared with placebo-treated controls (Girão *et al.*, 2003), in another pilot randomized clinical trial a significant *S. mutans* counts decrease was shown (Botelho *et al.*, 2007a).

The gel containing derivatives from both plants have shown more intensive antiinflammatory action than their isolated applications, inhibiting neutrophil infiltration as early as 6 h after its EPD induction as confirmed by the reduction of MPO activity on the gingival tissue. This effect was associated with a reduction in the chronic inflammatory cell influx on rat gingival tissue and periodontal ligament seen clearly from histopathology on day 11 after induction of EPD. These data are in accordance with the traditional use by rural populations (Monteiro *et al.*, 2006, 2007) and a previous report which demonstrated that the tannin fraction and chalcones extracted from the bark of *M. urundeuva* (Aroeira) inhibited direct and indirect neutrophil migration to the peritoneal cavity as well as vesical edema and increased vascular permeability in hemorrhagic cystitis induced by cyclophosphamide (Viana *et al.*, 1997). It has been also shown that the chalcone-enriched fraction isolated from *Aroeira* reduced neutrophil dependent paw edema induced by carrageenan (Viana *et al.*, 2003). Neutrophils are important cells in host defense against infecting bacteria. However, these cells have also been linked to tissue destruction. The importance of the acute neutrophil infiltration on gingival tissue in the evolution of periodontal disease has been demonstrated before (Liu and Pope, 2004; Menezes *et al.*, 2005). Thus, the reduction of neutrophil influx into gingival tissue as confirmed by MPO activity could be an explanation for the protective effect of the combined gel on the periodontal tissue including bone and cementum.

Effect of carvacrol (Car) and chalcones enriched fraction (CEF) on the alveolar bone loss of the experimental periodontal disease in rats

The treatment of animals subjected to 11 days of experimental periodontal disease with a dental gel based on a chalcones enriched fraction (CEF) and carvacrol (Car) reduced the alveolar bone loss. These changes reached statistical significance ($p < 0.05$) only in groups of animals receiving the DOX gel and Car + CEF combined gel, compared with the untreated animals subjected to EPD and those treated with vehicle gel (Fig. 2).

These data can be clearly seen in Fig. 3, which shows the macroscopic aspects of the contra lateral right side (unligated side) with no resorption of the alveolar bone, compared with the severe bone resorption with root exposure in the non-treated group (NT) (Fig. 3b) and vehicle based gel treated group. Figure 3c shows reduction in bone loss in animals subjected to experimental periodontitis and treated with the Car + CEF combined gel.

Effect of carvacrol (Car) and chalcones enriched fraction (CEF) on the histopathological analysis of EPD in rats

The histopathological analysis of the region between the first and second molars of the normal periodontium shows periodontal ligament (Pl), alveolar bone (Ab), cementum (Ce) and gingiva (g) (Fig. 3A; Table 1). The histopathology of the periodontium of the animals subjected to periodontitis that received vehicle gel (vehicle group) revealed intense inflammatory cell infiltration coupled with cementum destruction and complete alveolar process resorption (Table 1), receiving median score 3 (range, 2–3), similar to the periodontium of non-treated animals subjected to periodontitis which received median score 3 (range, 3–3) (Table 1; Fig. 3B). A reduction of inflammatory cell infiltration and a

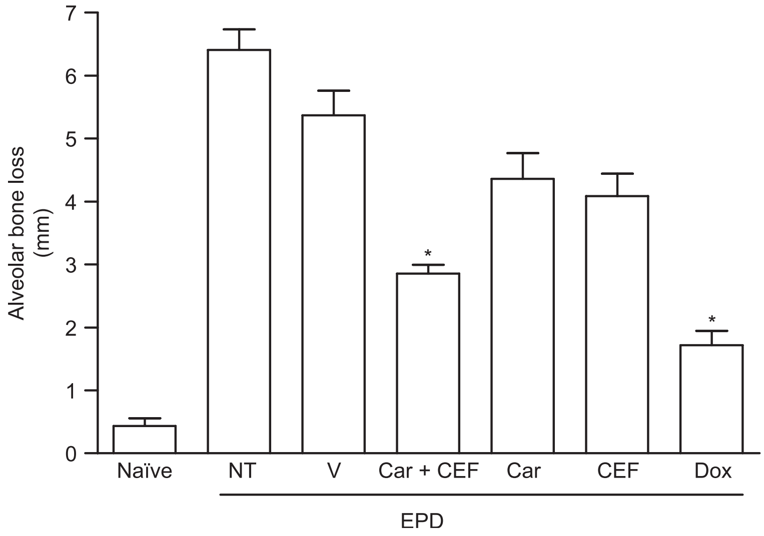

Figure 2. Effect of carvacrol (Car) and chalcones (CEF) on the alveolar bone loss in experimental periodontal disease (EPD) in rat: Vehicle gel (V), Car 300 µg/g based gel (Car), CEF 600 µg/g based gel (CEF), Doxycycline 100 µg/g based gel (Dox), or the combination of Car 300 µg/g and CEF 600 µg/g gel (Car + CEF) was administered topically, daily, in animals subjected to the EPD induction. EPD was also induced in non-treated animals (NT) and naïve group received no treatment and was not submitted to EPD induction. Bars represent the mean ± SEM of alveolar bone loss (mm; $n = 6$). Asterisk indicates a statistically significant difference between treated group and NT group (* $p < 0.05$; Bonferroni, ANOVA).

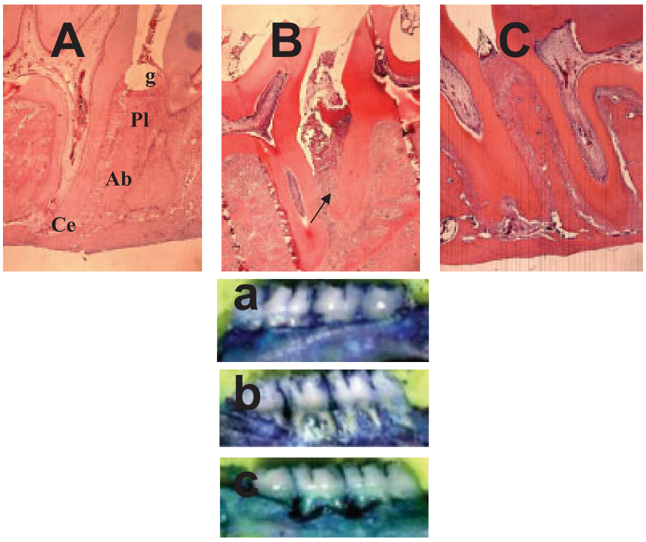

Figure 3. Histopathologic and macroscopic aspects of the effect of carvacrol (Car) and chalcones (CEF) on the experimental periodontal disease (EPD) in rat. Panel A: microscopic aspect of a normal periodontium of rat maxillae showing gingiva (g), periodontal ligament (Pl), cementum (Ce), alveolar bone (Ab); Panel B: periodontium of rat subjected to experimental periodontitis with no treatment showing severe inflammatory cell infiltration in both gingiva and periodontal ligament, cementum destruction (arrow) and total resorption of the alveolar process; Panel C: periodontium of rat subjected to experimental periodontitis and treated with Car + CEF gel showing absence of inflammatory cell infiltration and preservation of cementum and alveolar bone (hematoxylin and eosin stain; magnification ×40); Panel a: macroscopic aspects of a normal maxillae; panel b: macroscopic aspects of the maxillae excised from a non-treated animal subjected to experimental periodontitis showing severe bone resorption and root exposure; panel d: macroscopic aspects of the maxillae excised from an animal treated with Car + CEF showing significant reduction in bone resorption.

partial preservation of the cementum and of the alveolar process was found in the periodontium of animals subjected to experimental periodontitis and treated with carvacrol and chalcones (Car 300 µg/g + CEF 600 µg/g), (Fig. 3C; Table 1), receiving a median score 1 (range, 0–1) (Table 1). These values were statistically significant ($p < 0.05$), when compared with the non-treated and vehicle gel treated groups (Table 1).

Effect of carvacrol (Car) and chalcones enriched fraction (CEF) on the myeloperoxidase activity on gingiva of rats with EPD

The neutrophil infiltration was evaluated by the myeloperoxidase activity in the gingival tissue. A significant ($p < 0.05$) decrease in the myeloperoxidase activity in the gingival tissue was observed only in the group submitted to the Car + CEF combined based gel treatment, compared with the vehicle-treated rats (Fig. 4).

Antibacterial effect of carvacrol and chalcones enriched fraction (CEF) gel in oral microorganisms of rats with experimental periodontal disease

The microbiological analysis in the gingival tissue of rats subjected to the experimental periodontal disease showed that the Car and CEF gel treated group and the Car + CEF combined gel treated group had a significant ($p < 0.05$) decrease of periodontal microorganism growth (Fig. 5).

Effect of carvacrol (Car) and chalcones enriched fraction (CEF) on the weight loss

The weight loss, which is characteristically seen in the animals subjected to the experimental periodontitis (Lima *et al.*, 2000), was significantly ($p < 0.05$) prevented by the Car + CEF combined gel treatment and also by the carvacrol and chalcones gel treatment alone (Fig. 6).

Essential oil mouthwashes are available on the market for the treatment of oral infections. Thymol

Table 1. Histopathological analysis of the effect of carvacrol (Car) and chalcones (CEF) on experimental periodontal disease

Group	Naïve	NT	Vehicle	Car + CEF	Car	CEF	Doxycycline
Score	0 (0–0)	3 (3–3)[a]	3 (2–3)[a]	0 (0–1)[b]	1 (1–1)[a,b]	1 (1–1)[a,b]	0 (0–0)[b]

[a] $p < 0.05$ compared with naïve animals (Kruskal-Wallis).
[b] $p < 0.05$ compared with non-treated and vehicle-treated animals (Kruskal-Wallis).

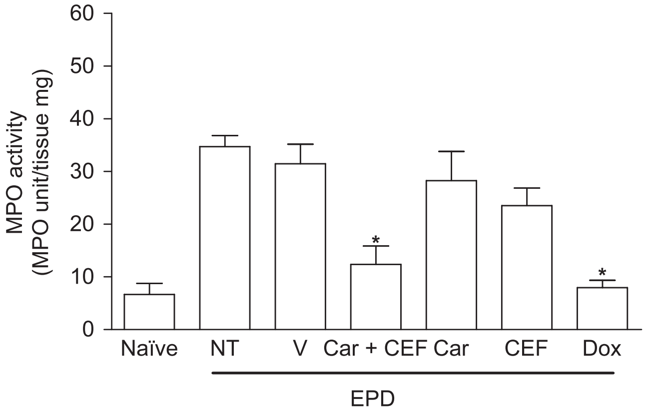

Figure 4. Effect of carvacrol (Car) and chalcones (CEF) on myeloperoxidase (MPO) activity in the maxillary gingival tissue of rats submitted to experimental periodontal disease (EPD). Vehicle gel (V), Car 300 μg/g based gel (Car), CEF 600 μg/g based gel (CEF) or the combination of Car 300 μg/g, Doxycycline 100 μg/g based gel (Dox), and CEF 600 μg/g gel (Car + CEF) was administered topically in animals subjected to EPD induction. EPD was also induced in non treated animals (NT) and naïve group received no treatment and was not submitted to EPD induction. Bars represent mean ± SEM of the activity of MPO/mg of tissue. * $p < 0.05$ was considered significantly different compared to NT group (ANOVA; Bonferroni's test).

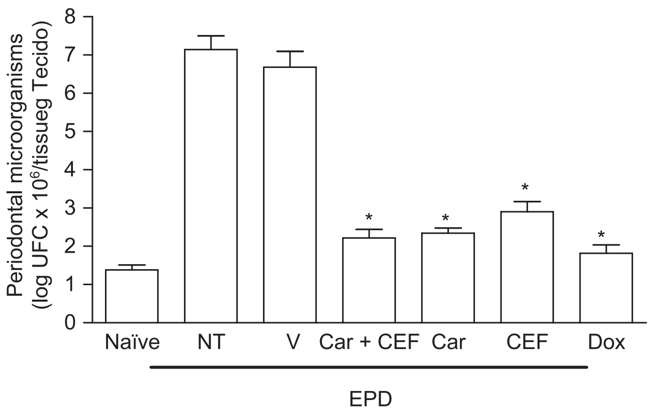

Figure 5. Antibacterial effect of carvacrol (Car) and chalcones (CEF) in bacteria involved in experimental periodontal disease (EPD) in rats. Vehicle gel (V), Car 300 μg/g based gel (Car) or CEF 600 μg/g based gel (CEF), Doxycycline 100 μg/g based gel (Dox) and the combination of Car 300 mg/g and CEF 600 mg/g gel (Car + CEF) was administered topically in animals submitted to the EPD. EPD was induced also in non-treated animals (NT). Results are reported as mean ± SEM of the colonies forming units (CFU) count/g of tissue from the rats subjected to EPD. Asterisk indicates a statistically significant difference between treated groups and NT group (* $p < 0.05$; Bonferroni, ANOVA).

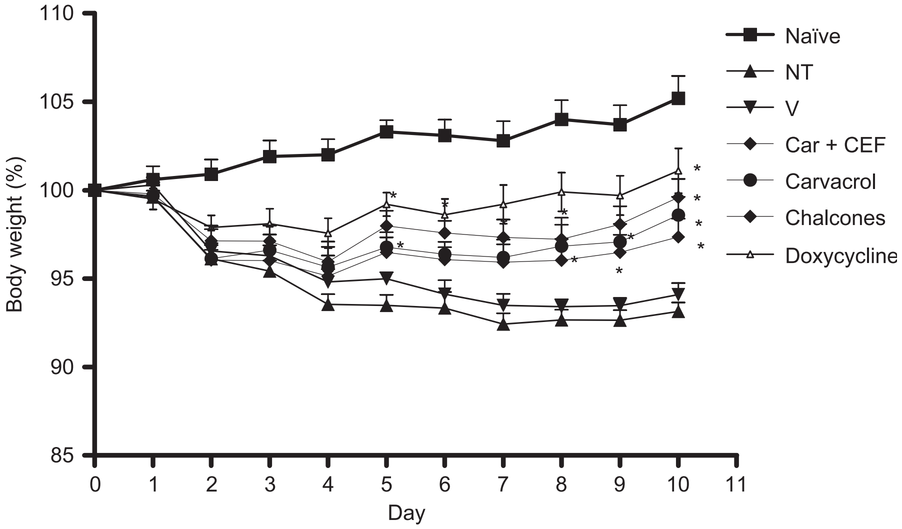

Figure 6. Effect of carvacrol (Car) 0.5% + chalcones (CEF) 5% on weight changes in rats submitted to the experimental periodontitis disease (EPD). Vehicle gel (V), Car 300 μg/g based gel (Car), CEF 600 μg/g based gel (CEF) or the combination of Car 300 μg/g, Doxycycline 100 μg/g based gel (Dox) and CEF 600 μg/g gel (Car + CEF) was administered topically in animals submitted to the EPD. EPD was induced also in non-treated animals (NT) and naïve group received no treatment and was not submitted to EPD induction. Data represent mean ± SEM ($n = 6$) of percentage (%) of body weight changes measured before and each day after the EPD induction until day 11. Asterisk indicates a statistically significant difference between treated groups and NT group (* $p < 0.05$; ANOVA, Bonferroni's test).

and carvacrol has shown good antibacterial efficacy against the microbes responsible for the periodontal disorders (Osawa *et al.*, 1990; Fine *et al.*, 2000; Botelho *et al.*, 2007b). Previous data have shown that a local application of a tea tree oil-containing essential oils in the form of gel showed greater efficacy on plaque and chronic gingivitis (Soukoulis and Hirsch, 2004), the same phenomenon was observed when a dental gel based on *M. urundeuva* and *Lippia sidoides* essential oil was locally applied into experimental periodontal groups of rats (Botelho *et al.*, 2007c). It is reasonable to speculate that therapeutic doses of the agent were delivered to bacteriane plaque at gingival surfaces for a longer period of time during the daily gel treatment (Pai *et al.*, 2004). Another main activity observed in the chalcones enriched fraction and carvacrol gels was the antimicrobial activity, confirming the use of this plant in Northeast Brazil in several types of infectious and inflammatory diseases (Lacoste *et al.*, 1996; Viana *et al.*, 2003; Monteiro *et al.*, 2007).

Bacterial plaque and its byproducts are considered the primary etiology of chronic gingival inflammation and periodontal diseases (Moore *et al.*, 1984). In accordance to previous studies, either the gel containing chalcones combined with carvacrol or the gels containing the chalcones or carvacrol presented significant antibacterial activity as shown by the periodontal bacteria count (Nowakowska, 2007; Botelho *et al.*, 2007c). Recently a study showed that the genus *Streptococcus* are involved in periodontal disease in rats (Klausen, 1991), and previous studies have demonstrated that carvacrol was effective against periodontal bacteria (Osawa *et al.*, 1990) and other genus of *Streptococci* involved in periodontal disorders (Botelho *et al.*, 2007b). A non-specific effect of local gel application can be disregarded by the fact that the control gel (vehicle) did not reduce the bacterial count. Thus, the inhibition of bacterial growth by CEF and/or carvacrol gel could be, at least partially, accounted for a reduction in MPO activity, bone and cementum destruction and could be due to a synergistic effect between chalcones and carvacrol on these parameters.

It is also reasonable to speculate that the anti-inflammatory effects shown by carvacrol and CEF gel could be partially due to an antioxidant property, known to be present in polyphenols, including chalcones and terpenoids (Souza *et al.*, 2007).

In conclusion, this study demonstrates that the local application of dental gel based on carvacrol (300 μg/g) from *Alecrim pimenta* and chalcones (600 μg/g) from *Aroeira-do-sertão* prevents alveolar bone resorption and has an antiinflammatory and antimicrobial effect in experimental periodontitis. The fact that these compounds 'in natura' are commercially used in gynecological and oral disorders in traditional medicine, highlight the need for further research in order to demonstrate their clinical effect for future inexpensive intervention in low-socioeconomic communities.

Acknowledgements

The authors thank Professor Telma L.G Lemos, providing carvacrol chemical isolate Professor Luewton Lemos from (Christhus College Research Center) and Professor Luciano Correa, (NEPI – Department of Community Health) for review on statistical data.

Maria Silvandira França Pinheiro, Department of Physiology and Pharmacology, and José Ivan Rodrigues de Sousa, Department of Morphology, Faculty of Medicine, Federal University of Ceará, Brazil, for technical assistance.

This work was supported by FUNCAP (9131/06) and CNPq the Brazilian Agencies for Scientific and Technological Development.

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